

**BOOK SHARING AND EXCHANGE
SYSTEM**

**AI23431- WEBTECHNOLOGY AND MOBILE
APPLICATION**

MINI PROJECT REPORT

Submitted by

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in partial fulfillment of the award of the degree of

**BACHELOR OF TECHNOLOGY
in
ARTIFICIAL INTELLIGENCE MACHINE LEARNING**



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING
RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI**

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CHENNAI - 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled "**BOOK SHARING AND EXCHANGE SYSTEM**" is the Bonafide work of JEEVAVENI S (241501076) and KIRUBASHINI R (241501087) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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EXTERNAL EXAMINER

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JEEVAVENI S (241501076)

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- To promote research activities amongst the students and the members of faculty that could benefit the society.
- To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

- PEO 1:** To equip students with essential background in computer science with emphasis on Artificial Intelligence, Machine Learning, basic electronics and applied mathematics.
- PEO 2:** To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications using emerging technologies.
- PEO 3:** To encourage research and innovative project development in the field of Artificial Intelligence, Machine Learning, Deep Learning, Networking, Security, Web development, Data Science and also emerging technologies for social benefit.
- PEO 4:** To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements

PROGRAMME OUTCOMES (POs)

- PO 1: Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO 2: Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO 3: Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO 4: Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- PO 5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO 6: The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO 7: Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental

contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate:

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational tasks, model real world problems using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVES

To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.

- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server-Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOMES

On completion of course students will be able to

CO 1: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.

CO 2: Design and implement dynamic web page with validation using JavaScript objects and by applying different event handling mechanism.

CO 3: Design and implement simple webpage to learn JSP and Servlet.

CO 4: Design and implement simple Application Design.

CO 5: Identify various concepts of mobile programming that make it unique from programming for other platforms

CO - PO — PSO matrices of course

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
AI23431.01	3	3	3	3	3	3	2	2	3	2	1	3	3	3	2
AI23431.02	3	3	3	3	3	3	2	2	3	2	1	1	3	3	2
AI23431.03	3	3	3	3	3	3	2	2	3	2	2	2	3	3	3
AI23431.04	3	3	3	3	3	3	2	2	3	2	2	3	3	3	3
AI23431.05	3	3	3	3	3	3	2	2	3	2	3	3	3	3	3
Average	3	3	3	3	3	3	2	2	3	2	1.8	2.1	3	3	2.4

Note: Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	The course provides fundamental knowledge in web and mobile application development, enabling students to gain practical technical skills, improve problem-solving abilities, and become industry-ready professionals.
SDG-8	Decent Work and Economic Growth	The subject equips students with job-oriented skills such as web development, client-side and server-side scripting, and mobile application design, increasing employability and supporting economic growth.
SDG-9	Industry, Innovation and Infrastructure	The course promotes innovation by teaching modern technologies used in developing web and mobile applications, contributing to digital infrastructure and technological advancement.
SDG-10	Reduced Inequalities	By enabling the development of accessible web and mobile applications, the course helps bridge the digital divide and ensures equal access to digital services for diverse users.

ABSTRACT

The Book Sharing and Exchange System (BSES) is a web-based platform designed to facilitate the sharing, borrowing, and exchanging of books among members of educational institutions and communities. With the growing need for affordable access to reading materials and efficient library management, the system provides a comprehensive digital solution to replace manual book tracking methods with an automated, secure, and user-friendly platform.

The application enables librarians and administrators to manage book catalogues electronically while allowing members to browse, borrow, return, and reserve books through a dedicated dashboard. By integrating role-based authentication, the system provides secure login access for both administrators and members, ensuring that book data and user information are protected and accessible only to authorized users.

The system includes features such as HTTP-based book management (GET, POST, PUT, DELETE), session management for tracking user authentication state, cookie-based preference storage, real-time book exchange listings, and a member management interface. Members can list books they wish to offer for free or exchange, while others can browse and request available books.

The architecture of the system is designed to be efficient and user-friendly, built using HTML, CSS, and JavaScript for the frontend with Node.js handling backend operations and MySQL for centralized data storage. The system demonstrates practical implementation of web technologies including HTTP methods, session handling, and cookie management. Overall, the Book Sharing and Exchange System improves accessibility to reading resources, reduces administrative workload, and promotes a culture of knowledge sharing within educational communities.

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LIST OF ABBREVIATIONS

BSES– Book Sharing and Exchange System

DB– Database

SQL– Structured Query Language

API– Application Programming Interface

UI– User Interface

HTML– HyperText Markup Language

CSS– Cascading Style Sheets

JS– JavaScript

HTTP– HyperText Transfer Protocol

REST– Representational State Transfer

CRUD– Create, Read, Update, Delete

ISBN– International Standard Book Number

SMTP– Simple Mail Transfer Protocol

HMAC– Hash-based Message Authentication Code

DRM– Digital Rights Management

CHAPTER 1

INTRODUCTION

1.1 General

Book management plays a vital role in educational institutions as it facilitates access to knowledge resources and promotes academic learning. Traditionally, books are managed using manual registers and physical inventory systems, which are time-consuming and prone to errors. Manual library systems also make it difficult to track book availability, manage borrowing records efficiently, and facilitate peer-to-peer book exchange among students and faculty.

The Book Sharing and Exchange System is designed to automate the process of managing and sharing books using a digital web-based platform. The system allows administrators and librarians to manage book catalogues electronically while enabling members to browse available books, borrow, return, and exchange reading materials through a secure dashboard. By maintaining book data in a centralized database, the system ensures accuracy, easy retrieval of records, and improved access to library resources.

This system also reduces administrative workload for librarians and provides transparency for members regarding book availability. With features such as genre-based book filtering, HTTP method demonstrations, session management, cookie-based preferences, and a book exchange marketplace, the system modernizes library operations in educational institutions.

1.2 Need for Book Sharing and Exchange System

Many educational institutions still rely on traditional library management approaches, which lead to challenges such as inaccurate inventory records, difficulty tracking borrowed books, lack of transparency for members, and no platform for peer-to-peer book sharing. Manual systems require significant effort to maintain records of borrowed books, calculate due dates, and notify members about overdue returns.

A digital book sharing and exchange system helps overcome these limitations by providing an automated and efficient platform. Librarians can update the catalogue quickly, and data is stored securely in a database. Members can also check book availability in real time, request exchanges with other users, and receive alerts about overdue items. The system promotes a reading culture by making book access easier and more transparent.

1.3 Scope of the System

The Book Sharing and Exchange System is developed to simplify the process of cataloguing, lending, and exchanging books in educational institutions. The system provides role-based access for administrators/librarians and regular members through secure authentication.

Administrators can add, update, and remove books from the catalogue, manage member accounts, and monitor borrowing activity. Members can log in to browse the book catalogue, borrow and return books, reserve titles, and participate in the book exchange marketplace. The system also maintains session data for active users and stores preferences using browser cookies.

The scope of the system also includes demonstrating web technology concepts such as HTTP GET, POST, PUT, and DELETE methods through live API interactions, HTTP session management with activity logging, and cookie preference management. It can be deployed in schools, colleges, community libraries, and other educational institutions.

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Book Sharing and Exchange System.

Chapter 1 provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2 presents the literature survey discussing existing library and book management systems and related work.

Chapter 3 explains the problem formulation and objectives of the proposed system.

Chapter 4 describes the system design, including architecture and system requirements.

Chapter 5 discusses the implementation of the system and explains the different modules with screenshots.

Chapter 6 presents the results and performance evaluation of the system.

Chapter 7 concludes the project and discusses possible future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Library management and book sharing are essential activities in educational institutions. Traditional approaches involve physical catalogues and manual registers, which are time-consuming and inefficient. With the advancement of web technologies, several digital systems have been developed to improve efficiency, accuracy, and accessibility in book management and lending operations.

Recent work highlights the growing adoption of web-based library systems, RESTful APIs, session-based authentication, and community book exchange platforms. These systems aim to reduce manual workload, improve data accuracy, and provide real-time access to library resources for both administrators and users.

2.2 Existing Book Management Systems

Many educational institutions have adopted digital library management systems to replace traditional paper-based methods. Some systems use barcode scanning or RFID technology to track books as they are borrowed and returned, reducing errors and speeding up the checkout process.

Web-based library systems have become increasingly popular as they allow users to search the catalogue and manage their accounts from any device. These platforms typically provide separate dashboards for librarians and members, enabling efficient management of borrowing records and notifications for overdue books.

Community-driven book exchange platforms allow users to list books they no longer need and request books from other members. These systems promote resource sharing and reduce the cost of accessing reading materials for students and researchers.

2.3 Related Work

Several research studies have proposed approaches to automate library management and book sharing. Studies on REST API-based library systems demonstrate how HTTP methods (GET, POST, PUT, DELETE) can be used to build scalable and maintainable backend services for managing book catalogues and member records.

Research on session management in web applications highlights the importance of server-side session storage for maintaining user authentication state across multiple page visits. Cookie-based preference management has also been studied as a method for persisting user settings such as theme preferences and reading history across sessions.

2.4 Summary

From the literature survey, it is clear that digital book sharing and library management systems provide significant advantages over traditional manual methods. They improve accuracy, reduce time consumption, and enable centralized data management accessible from any location. A web-based Book Sharing and Exchange System built with HTML, CSS, JavaScript, Node.js, and MySQL can provide a comprehensive and cost-effective solution that covers book catalogue management, member management, HTTP method demonstrations, session handling, and a peer-to-peer exchange marketplace.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In many educational institutions, book lending and sharing is still managed using paper-based registers and spreadsheets. This traditional approach is time-consuming and leads to several problems including inaccurate inventory records, difficulty in tracking borrowed books and due dates, lack of transparency for members about book availability, and no formal mechanism for peer-to-peer book exchange.

Members often do not have easy access to the library catalogue online, which makes it difficult to check availability before visiting in person. Additionally, there is no existing platform for students and faculty to share or donate books they no longer need with others who could benefit from them.

Therefore, there is a need for a web-based system that can digitize the book management process, provide real-time catalogue access, support book exchange between members, and demonstrate key web technology concepts including HTTP methods, session management, and cookie-based preferences.

3.2 Objectives of the System

The main objective of the Book Sharing and Exchange System is to develop a digital web platform that simplifies the process of managing library books and facilitating peer-to-peer book exchange. The specific objectives include:

- To maintain a centralized book catalogue with real-time status updates.
- To provide secure role-based login access for administrators and members.
- To allow members to borrow, return, and reserve books online.
- To enable users to list and request books through a book exchange marketplace.
- To demonstrate HTTP GET, POST, PUT, and DELETE methods through live API interactions.
- To implement HTTP session management for tracking user authentication state.
- To use cookies for storing user preferences and session data.
- To reduce manual errors and administrative workload in library operations.
- To improve transparency and accessibility of book availability information.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, and maintainability.

The Book Sharing and Exchange System is designed as a web-based application that allows administrators to manage the book catalogue and members to browse and borrow books. The system uses a client-server architecture where users interact through a web browser interface while the backend processes requests and stores data in a MySQL database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

Presentation Layer

This layer represents the user interface where administrators and members interact with the system. It includes web pages such as the login page, library dashboard, HTTP methods demo, session management page, cookies preferences page, book catalogue, book exchange, member management, and user profile. These are developed using HTML, CSS, and JavaScript.

Application Layer

This layer handles the business logic of the system. It processes user requests, verifies login credentials, manages book records, handles HTTP API requests, maintains session data, and communicates with the database. The backend is implemented using Node.js.

Database Layer

This layer stores all system data including book catalogue information, member records, borrowing history, exchange listings, and session data. The MySQL database ensures secure and efficient storage and retrieval of all information.

4.3 Database Design

The database is structured to support all system functionalities efficiently. Key tables include: Books (book details, status, copies), Members (member profiles and credentials), BorrowingRecords (borrowing and return history), ExchangeListings (peer-to-peer book offerings), and Sessions (active user sessions and activity logs).

4.4 System Requirements

4.4.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.4.2 Software Requirements

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Book Sharing and Exchange System is implemented using a web-based approach that allows users to access the system through any modern browser. The development follows a modular approach where each functionality is built as a separate module, improving maintainability and allowing future enhancements.

The frontend is developed using HTML, CSS, and JavaScript providing an intuitive and visually appealing interface with a warm library-themed design. The backend is implemented using Node.js, handling authentication, book management, HTTP API processing, session tracking, and cookie management. MySQL stores all system data centrally.

5.2 Module Descriptions

The Book Sharing and Exchange System consists of several integrated modules that work together to deliver complete library management and book exchange functionality.

5.2.1 Login and Authentication Module

The Login and Authentication Module serves as the secure entry point to the system. Users select their role (Admin/Librarian or Member) and provide their credentials to gain access to the appropriate dashboard. Role-based authentication ensures that only authorized users can access administrative features.

The login page displays the system title "Book Sharing & Exchange System" with the subtitle "Library Management Platform." Demo credentials are provided for testing purposes.

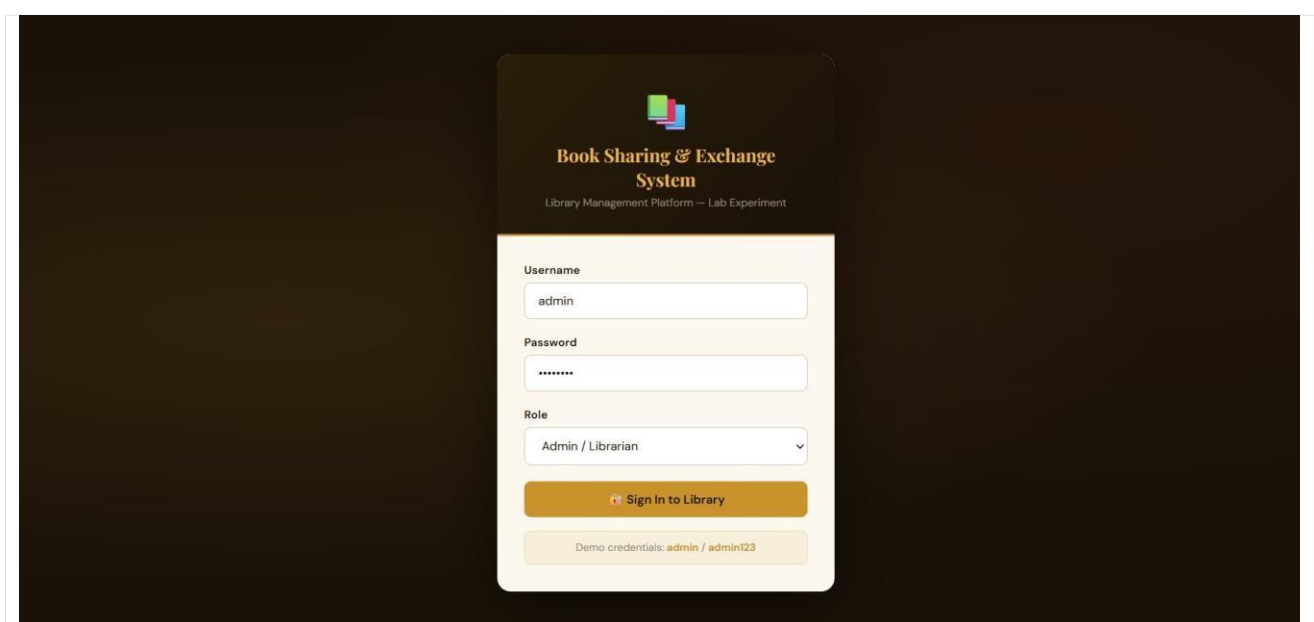


Fig 5.1 - Login Page (Role-based Authentication)

5.2.2 Library Dashboard Module

The Library Dashboard provides a comprehensive overview of the book sharing and exchange system. After login, administrators can see key statistics including total books in the catalogue, active members, books currently borrowed, and overdue returns. The dashboard also shows a recent activity feed with real-time transaction records and a notifications panel for important alerts such as overdue books, new donations, and system maintenance updates.

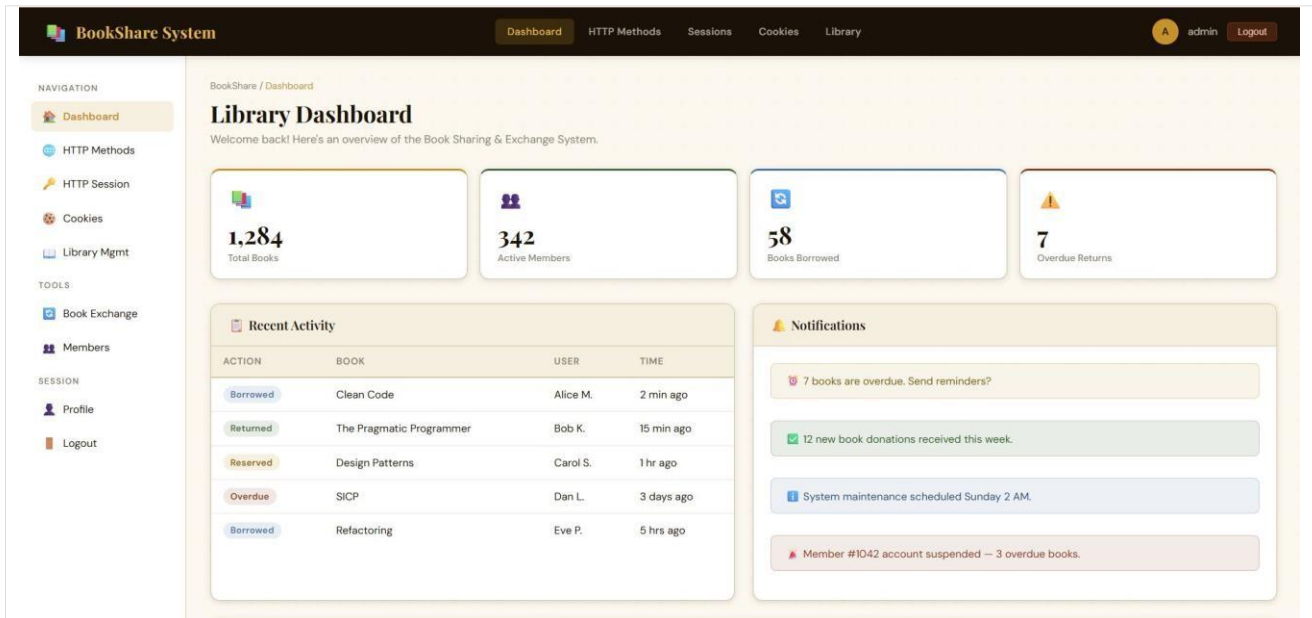


Fig 5.2 - Library Dashboard with Statistics and Activity Feed

5.2.3 HTTP Methods Module (GET & POST)

This module demonstrates the practical implementation of HTTP methods used in the BookShare REST API. The GET section allows users to send retrieval requests to fetch the book list from the server by search query. The POST section provides a form to add new books to the catalogue by submitting book data to the server endpoint. Both sections display the actual request endpoint and the live server response in real time.

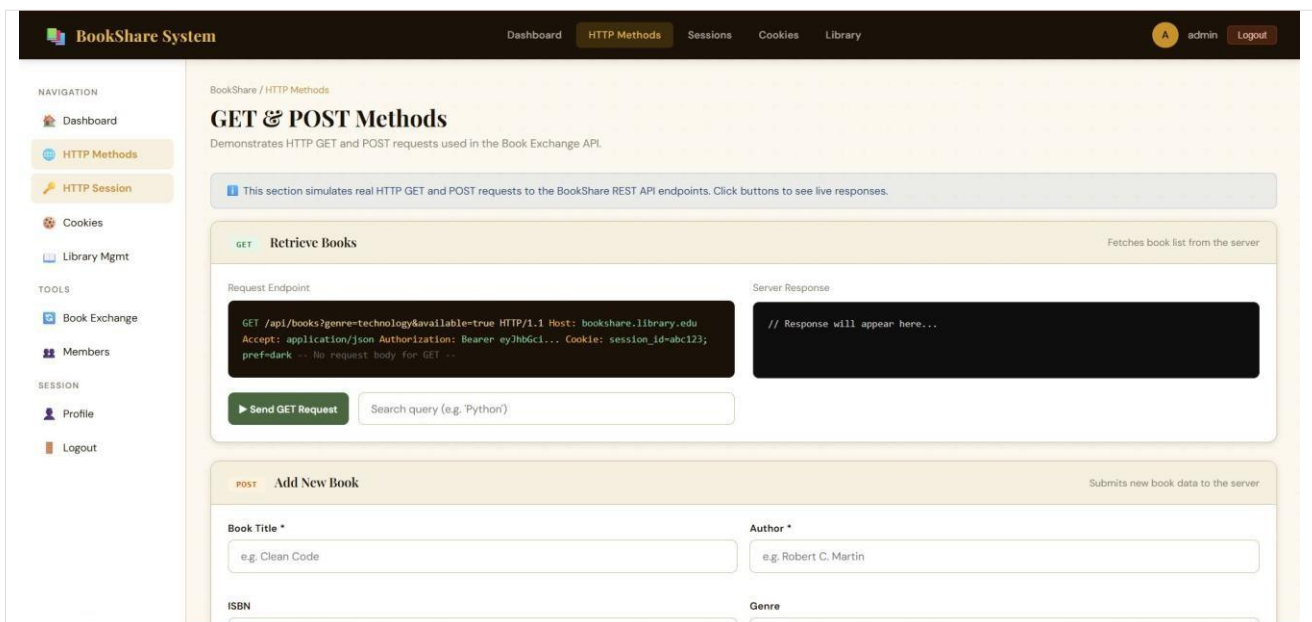


Fig 5.3 - HTTP GET and POST Methods for Book Management

5.2.4 HTTP Methods Module (PUT & DELETE)

Continuing the HTTP methods demonstration, this section covers the PUT method for updating existing book records by specifying a Book ID and the field to modify (such as status, availability, or location). The DELETE section allows removal of books from the catalogue by providing the Book ID and a reason (such as Lost/Damaged or Withdrawn). Both operations display live responses in the interface.

The screenshot displays the 'BookShare System' interface with a dark header bar containing navigation links: Dashboard, HTTP Methods, Sessions, Cookies, and Library. The user is logged in as 'admin'. The main content area is divided into two panels. The left panel, titled 'PUT Update Book', contains a 'Book ID' field (e.g., BK-1042), a 'Field to Update' dropdown (set to 'Status'), and a 'New Value' field (e.g., 'New value...'). Below these fields are buttons for 'Send PUT Request' and a status bar showing '// Awaiting...'. The right panel, titled 'DELETE Remove Book', contains a 'Book ID to Delete' field (e.g., BK-1042) and a 'Reason' dropdown (set to 'Lost / Damaged'). Below these fields are buttons for 'Send DELETE Request' and a status bar showing '// Awaiting...'. A 'Send POST Request' button is visible at the top of the main content area.

Fig 5.4 - HTTP PUT and DELETE Methods for Book Updates and Removal

5.2.5 HTTP Session Management Module

The HTTP Session Management module tracks user authentication state across the application. It displays active session data including the Session ID, Username, Role, Login Time, Expiry Time, and IP Address. Session variables are shown in JSON format for transparency. The module also provides controls to refresh or invalidate the current session and displays a Session Activity Log showing all navigation and login events during the current session.

The screenshot displays the 'BookShare System' interface with a dark header bar containing navigation links: Dashboard, HTTP Methods, Sessions, Cookies, and Library. The user is logged in as 'admin'. The main content area is titled 'HTTP Session Management' and includes a subtitle 'Tracks user authentication state, session tokens, and activity log.' The interface is divided into three main sections. The top section, 'Active Session Data', displays session details in a grid: SESSION ID (sess_sxk3xuw2ep), USERNAME (admin), ROLE (admin), LOGIN TIME (5:45:18 PM), EXPIRES AT (6:15:18 PM), and IP ADDRESS (192.168.1.102). Below this grid are buttons for 'Refresh Session' and 'Invalidate Session'. The middle section, 'Session Variables', displays a JSON object: { "session_id": "sess_sxk3xuw2ep", "username": "admin", "role": "admin", "login_time": "2026-04-22T12:15:18.970Z", "expires_at": "2026-04-22T12:45:18.971Z", "permissions": ["read", "write", "delete", "admin"], "cart": [], "last_page": "dashboard", "theme": "light" }. The bottom section, 'Session Activity Log', displays a log entry: '5:46:17 PM Navigated to session action'. A left sidebar contains navigation links: Dashboard, HTTP Methods, HTTP Session (highlighted), Cookies, Library Mgmt, Tools (Book Exchange, Members), and Session (Profile, Logout).

Fig 5.5 - HTTP Session Management with Active Session Data

5.2.6 Session Activity Log and Configuration

The lower section of the Session Management module displays a chronological activity log recording all user actions such as login events, page navigations, and system interactions with timestamps. The Session Configuration panel allows administrators to set session timeout duration, maximum concurrent sessions, storage type (server-side), and the token algorithm (HS256 HMAC) used for session security.

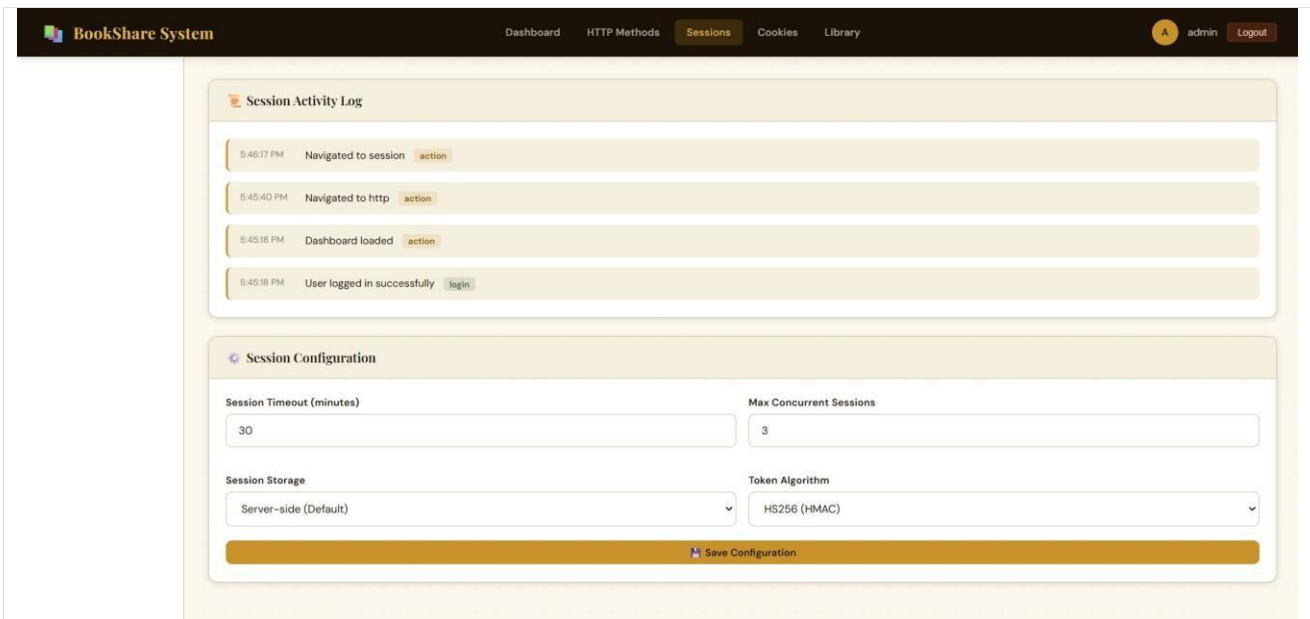


Fig 5.6 - Session Activity Log and Session Configuration Settings

5.2.7 Cookie Preferences Module

The Cookie Preferences module manages browser cookies used by the Book Sharing and Exchange System. It displays all currently active cookies including SESSION_ID, USERNAME, ROLE, THEME, and LAST_LOGIN, along with their expiry information and security settings. Users can manage cookie preferences by enabling or disabling different cookie categories: Essential Cookies (required for login), Preference Cookies (UI settings), Analytics Cookies (usage data), Reading History (recommendation tracking), and Third-Party Cookies.

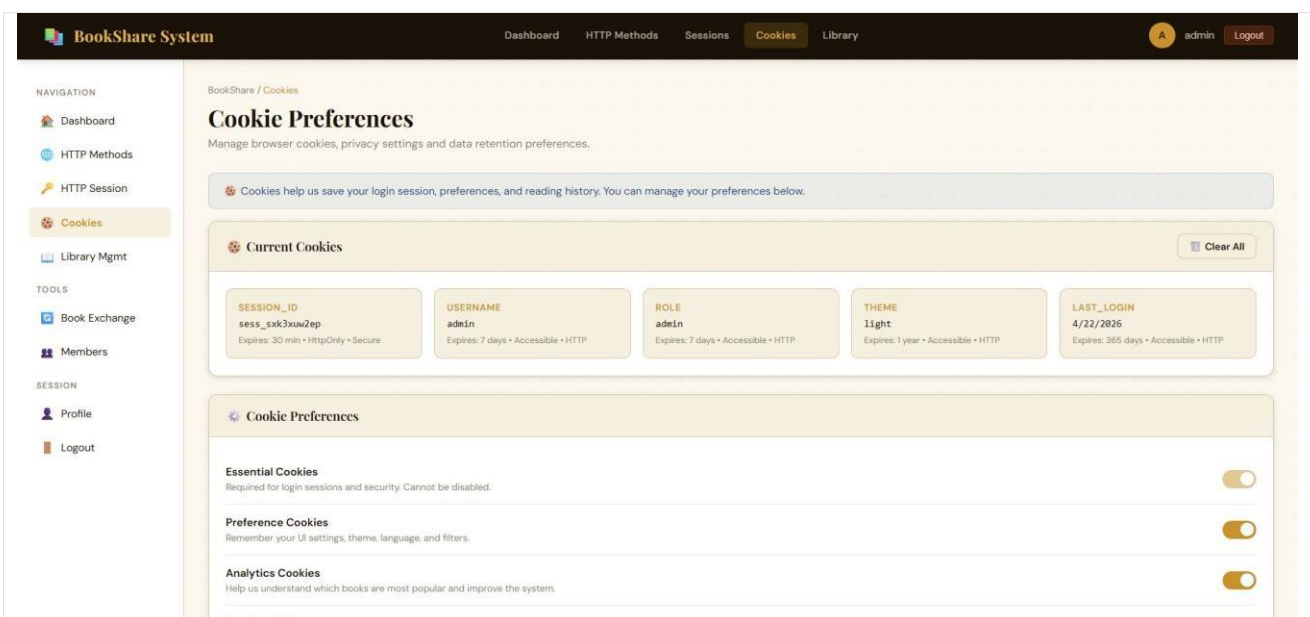


Fig 5.7 - Cookie Preferences and Current Active Cookies

5.2.8 Custom Cookie Management

The Cookie Management module also includes a Set Custom Cookie section where administrators can create custom cookies by specifying the cookie name, value, expiry duration in days, and security settings (Secure + HttpOnly). Options are also available to Accept All or Reject Optional cookies. This demonstrates the complete cookie lifecycle management in a web application.

Fig 5.8 - Custom Cookie Creation and Cookie Preference Controls

5.2.9 Library Management Module

The Library Management module is the core of the system, providing a complete book catalogue interface. Administrators can search books by title, author, or ISBN, filter by genre, and view all books with their current status (Available, Borrowed, Reserved, or Overdue). Each book entry shows the Book ID, Title, Author, Genre, ISBN, number of copies, and available actions (Issue or View). New books can be added using the "+ Add Book" button.

#	TITLE	AUTHOR	GENRE	ISBN	COPIES	STATUS	ACTIONS
BK-001	Clean Code	Robert C. Martin	Technology	978-0132350884	3	Available	Issue View
BK-002	The Pragmatic Programmer	Hunt & Thomas	Technology	978-0135957859	2	Borrowed	Issue View
BK-003	Design Patterns	Gang of Four	Technology	978-0201633610	1	Reserved	Issue View
BK-004	SICP	Abelson & Sussman	Technology	978-0262510875	1	Overdue	Issue View
BK-005	Refactoring	Martin Fowler	Technology	978-0134757599	4	Available	Issue View
BK-006	The Alchemist	Paulo Coelho	Fiction	978-0062315887	5	Available	Issue View

Fig 5.9 - Library Management with Book Catalogue and Status Tracking

5.2.10 Book Exchange Module

The Book Exchange module enables peer-to-peer book sharing among members. Users can offer books by listing the title, condition, exchange type (Free Donation or Exchange), and a description. On the right side, current book offerings from other members are displayed, showing the book title, author, condition, and exchange type. Members can request a listed book or view its details.

BookShare System | Dashboard | HTTP Methods | Sessions | Cookies | Library | admin | Logout

NAVIGATION

- Dashboard
- HTTP Methods
- HTTP Session
- Cookies
- Library Mgmt

TOOLS

- Book Exchange
- Members

SESSION

- Profile
- Logout

BookShare / Book Exchange

Book Exchange

Post books for sharing or browse books offered by other members.

Offer a Book

Book Title
Enter title...

Condition
Excellent

Exchange Type
Free Donation

Description
Describe the book...

List Book

Browse Offerings

JavaScript: The Good Parts
by Douglas Crockford • Good condition

Free

Request View Details

You Don't Know JS
by Kyle Simpson • Excellent condition

Exchange

Request View Details

Fig 5.10 - Book Exchange Marketplace for Peer-to-Peer Book Sharing

5.2.11 Member Management Module

The Member Management module allows administrators to view and manage all registered library members. The interface displays a table showing Member ID, Name, Email, number of books borrowed, join date, and account status (Active or Suspended). Administrators can add new members using the "+ Add Member" button and view individual member profiles. Suspended accounts (such as those with overdue books) are clearly marked.

BookShare System | Dashboard | HTTP Methods | Sessions | Cookies | Library | admin | Logout

NAVIGATION

- Dashboard
- HTTP Methods
- HTTP Session
- Cookies
- Library Mgmt

TOOLS

- Book Exchange
- Members

SESSION

- Profile
- Logout

BookShare / Members

Member Management

Registered Members + Add Member

ID	NAME	EMAIL	BOOKS BORROWED	JOINED	STATUS	ACTIONS
MEM-001	Alice Murugan	alice@mail.com	3	Jan 2024	Active	View
MEM-002	Bob Kumar	bob@mail.com	1	Mar 2024	Active	View
MEM-003	Carol Selvam	carol@mail.com	5	Feb 2023	Suspended	View

Fig 5.11 - Member Management with Registration and Status Tracking

5.2.12 User Profile Module

The User Profile module allows users to view and edit their personal information. The profile page displays the user avatar, full name, role, and membership date. Users can update their full name, email address, phone number, and department through an edit form. A Change Password section allows updating credentials by entering the current password and a new password. Changes are saved using the Save Changes button.

The screenshot displays the 'User Profile' management interface within the 'BookShare System'. The top navigation bar includes links for 'Dashboard', 'HTTP Methods', 'Sessions', 'Cookies', and 'Library', along with a user profile icon labeled 'admin' and a 'Logout' button. The left sidebar contains a 'NAVIGATION' menu with 'Dashboard', 'HTTP Methods', 'HTTP Session', 'Cookies', and 'Library Mgmt', followed by a 'TOOLS' section with 'Book Exchange' and 'Members', and a 'SESSION' section with 'Profile' (highlighted) and 'Logout'. The main content area is titled 'User Profile' and features a user card for 'Admin User' (role: Admin) with a membership date of 'January 2024'. To the right is an 'Edit Profile' form with fields for 'Full Name' (Admin User), 'Email' (admin@library.edu), 'Phone' (+91 98765 43210), and 'Department' (Library Administration). Below these is a 'Change Password' section with 'Current Password' and 'New Password' fields. A prominent orange 'Save Changes' button is at the bottom of the form.

Fig 5.12 - User Profile Management Page

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Book Sharing and Exchange System was successfully developed and tested using a web-based platform. The system allows administrators to manage the complete book catalogue electronically while enabling members to browse books, borrow, return, reserve, and exchange reading materials through a secure dashboard.

During testing, the system performed efficiently in handling user login with role-based access, book catalogue operations using HTTP methods (GET, POST, PUT, DELETE), session tracking with activity logging, cookie-based preference management, peer-to-peer book exchange listings, and member management. The system correctly processed all test cases and stored data accurately in the MySQL database.

The system also successfully demonstrated key web technology concepts including live HTTP API interactions, session management with expiry and invalidation controls, and cookie lifecycle management including creation, preference toggling, and deletion.

6.2 Performance Evaluation

The performance of the Book Sharing and Exchange System was evaluated based on usability, accuracy, and response time. The system provides a simple and visually appealing interface themed around a warm library aesthetic that makes it easy for both administrators and members to interact with the application.

The book catalogue data is maintained in a centralized MySQL database, ensuring records are accurate and retrievable quickly. The use of Node.js for backend processing allows the system to handle simultaneous user requests efficiently. Testing confirmed that the system performs well under normal usage conditions and provides reliable results for all implemented features.

6.3 System Testing

The system was tested using a series of test cases covering all major modules:

- Test Case 1: User Login with Valid Credentials - PASS
- Test Case 2: Role-Based Access Control (Admin vs Member) - PASS
- Test Case 3: HTTP GET Request for Book Retrieval - PASS
- Test Case 4: HTTP POST Request to Add New Book - PASS
- Test Case 5: HTTP PUT Request to Update Book Status - PASS
- Test Case 6: HTTP DELETE Request to Remove Book - PASS
- Test Case 7: Session Creation and Activity Logging - PASS
- Test Case 8: Session Refresh and Invalidation - PASS
- Test Case 9: Cookie Preference Save and Retrieval - PASS
- Test Case 10: Book Exchange Listing and Request - PASS
- Test Case 11: Member Management (Add/View/Suspend) - PASS
- Test Case 12: User Profile Update - PASS

All test cases passed successfully, confirming that the system is functioning correctly and is ready for deployment in an educational environment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Book Sharing and Exchange System was developed to provide an efficient and reliable digital platform for managing library books and facilitating peer-to-peer book sharing in educational institutions. The system replaces traditional manual library management methods with a modern web-based solution that simplifies cataloguing, lending, returning, and exchanging books.

The application allows librarians and administrators to manage the book catalogue efficiently while enabling members to access library services through a secure, role-based login system. By maintaining all data in a centralized MySQL database, the system ensures accuracy, transparency, and quick retrieval of information whenever required.

The system also successfully demonstrates practical web technology concepts including HTTP REST API methods, server-side session management, and browser cookie handling. The implementation reduces manual workload, minimizes errors in library records, and improves overall efficiency. The web-based design makes the system accessible and easy to use from any device with a browser.

7.2 Future Enhancements

Although the current system provides comprehensive features for book management and exchange, several improvements can be made in the future to enhance its functionality:

- Integration of barcode or QR code scanning for faster book checkout and return processing.
- Development of a mobile application for iOS and Android to allow members to access library services from smartphones.
- Implementation of automated email or SMS notifications for overdue book reminders and new book arrival alerts.
- Addition of a book recommendation engine using machine learning to suggest books based on borrowing history.
- Integration with external digital library databases for expanded catalogue access.
- Support for e-book lending with digital rights management (DRM) for digital reading materials.

These enhancements will make the Book Sharing and Exchange System more advanced, user-friendly, and suitable for large-scale institutional deployment.

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APPENDIX

Appendix 1: System Pseudocode

System Initialization Logic

```
START
  Connect to MySQL database
  IF database connection is successful
    Start server on port 3000
    Load system homepage (index.html)
  ELSE
    Display database connection error
  END IF
END
```

User Login Logic

```
FUNCTION Login(username, password, role)
  IF role = admin
    Check admin table for matching credentials
  ELSE IF role = member
    Check members table for matching credentials
  END IF
  IF credentials are valid
    Create user session with session_id
    Store session variables (username, role, login_time)
    Set authentication cookies
    Redirect to corresponding dashboard
  ELSE
    Display "Invalid login credentials"
  END IF
END FUNCTION
```

Book Management Logic (HTTP Methods)

```
FUNCTION GetBooks(query, genre)
  SELECT * FROM books WHERE title LIKE query AND genre = genre
  Return book list as JSON
END FUNCTION

FUNCTION AddBook(title, author, isbn, genre, copies)
  INSERT INTO books VALUES (title, author, isbn, genre, copies, "Available")
  Return success message
END FUNCTION

FUNCTION UpdateBook(book_id, field, value)
  UPDATE books SET field = value WHERE id = book_id
  Return updated record
END FUNCTION

FUNCTION DeleteBook(book_id, reason)
  DELETE FROM books WHERE id = book_id
  Log deletion with reason
  Return confirmation
END FUNCTION
```

Session Management Logic

```
FUNCTION CreateSession(user_id, role)
  session_id = GenerateUniqueToken()
  Store in sessions table:
    session_id, user_id, role, login_time, expiry_time, ip_address
  Set session cookie
END FUNCTION
```

```
Return session_id
END FUNCTION

FUNCTION ValidateSession(session_id)
  IF session exists AND not expired
    Return session data
  ELSE
    Invalidate session
    Redirect to login
  END IF
END FUNCTION
```

Book Exchange Logic

```
FUNCTION ListBookForExchange(user_id, title, condition, exchange_type, description)
  INSERT INTO exchange_listings VALUES (...)
  Return listing_id
END FUNCTION

FUNCTION RequestBook(listing_id, requester_id)
  UPDATE exchange_listings SET status = "Requested" WHERE id = listing_id
  Notify book owner
  Return confirmation
END FUNCTION
```

Appendix 2: System Test Results Summary

Test Case	Description	Result
TC-01	User Login with Valid Credentials	PASS
TC-02	Role-Based Access Control	PASS
TC-03	HTTP GET - Retrieve Books	PASS
TC-04	HTTP POST - Add New Book	PASS
TC-05	HTTP PUT - Update Book Status	PASS
TC-06	HTTP DELETE - Remove Book	PASS
TC-07	Session Creation and Logging	PASS
TC-08	Session Refresh and Invalidation	PASS
TC-09	Cookie Preference Management	PASS
TC-10	Book Exchange Listing	PASS
TC-11	Member Management	PASS
TC-12	User Profile Update	PASS